

# M3U Web Picker

## Step-by-Step User Guide

v22.1-rc2 - Beginner-first, top-to-bottom guide with edge cases and troubleshooting

### Who this guide is for

Someone who has an IPTV provider and wants a manageable channel lineup and TV guide in Jellyfin, but does not necessarily know or care what M3U, XMLTV, Xstream, canonical event IDs, or EPG reconciliation mean.

### RC2 tester note

This guide describes v22.1-rc2 as it exists now. RC2 currently exposes /playlist/custom.m3u and /epg/combined.xml. The planned next cleanup is to rename the public-facing outputs to /playlist/channels.m3u and /epg/epg.xml and keep the old routes as compatibility aliases. Always trust the Copy buttons shown by the running app.

### Known RC2 issue

Replay suppression is not fully correct yet. In current testing, late-night / midnight Phillies rebroadcasts can still appear even when replays are disabled. That is a known RC2 bug, not the intended behavior.

## The whole app in one sentence

Give M3U Web Picker your IPTV source, choose the normal channels you want, optionally describe the sports you care about, and it publishes one playlist plus one TV guide for your media server.

## The beginner path

1. Install Docker and Git.
2. Start M3U Web Picker and open the web page.
3. Load one Primary IPTV provider.
4. Pick the normal channels you want to keep.
5. Leave the United States public EPG enabled (or choose your countries).
6. Optionally enable Sports Automation and add sport / league / team selections.
7. Optionally configure the supported sports schedule API for better matching.
8. Run Update Now and wait for the full update to finish.
9. Add the app playlist and EPG URLs to Jellyfin.

## Guide map - follows the page from top to bottom

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## 0. Environment Setup

This section gets the application running before you touch IPTV configuration. If you already have the RC2 container running on port 9999, you can skip to Section 1.

### 0.1 What you need

- Windows 10/11, macOS, or a modern Linux distribution.
- Docker Desktop (Windows/macOS) or Docker Engine plus the Docker Compose plugin (Linux).
- Git, if you are installing from GitHub.
- A modern web browser.
- An IPTV provider or local M3U file if you want live channels. Sports Automation cannot invent playable streams by itself.
- For Sports Automation schedule assistance, an API-SPORTS account is optional. Provider-derived matching still works without it.

## You do NOT need a local Python setup

The normal installation runs inside Docker. You do not need to install Flask, SQLite, Python packages, or a web server directly on the host.

## 0.2 Verify Docker and Git

```
docker --version
docker compose version
git --version
```

If a command says 'not found', install Docker Desktop (Windows/macOS) or Docker Engine + Compose (Linux), and Git, then reopen the terminal. If Docker is installed but cannot connect, start Docker Desktop/the Docker service. Linux permission error: try `sudo`. Before continuing, `docker run --rm hello-world` should succeed.

## 0.3 Fresh install from GitHub

RC2 currently lives on the sports branch. These commands deliberately clone that branch. When the work is merged into the default branch, the branch option can be removed.

### macOS

```
cd ~/Desktop && git clone --branch sports --single-branch https://github.com/zschmook/m3u-web-picker.git && cd m3u-web-picker && mkdir -p backups && docker compose up -d --build
```

### Linux

```
cd ~ && git clone --branch sports --single-branch https://github.com/zschmook/m3u-web-picker.git && cd m3u-web-picker && mkdir -p backups && docker compose up -d --build
```

### Windows PowerShell

```
cd ([Environment]::GetFolderPath("Desktop")); git clone --branch sports --single-branch https://github.com/zschmook/m3u-web-picker.git; cd m3u-web-picker; New-Item -ItemType Directory -Force backups | Out-Null; docker compose up -d --build
```

### If the folder already exists

`git clone` will refuse to overwrite an existing `m3u-web-picker` folder. If that folder is already the correct repository, enter it and use `git status` first. If it is clean, `git pull` followed by `docker compose up -d --build` is the normal update path.

## 0.4 Open the application

```
http://localhost:9999
```

The normal Compose stack uses host port 9999. Port 10000 is reserved for the separate sports-debug Compose file used during development/testing; normal users should not use the debug stack.

## 0.5 If port 9999 is already in use

Create or edit a `.env` file next to `docker-compose.yml` and choose another host port:

```
M3U_HOST_PORT=10001
```

Then rebuild/recreate the container. The port on the right side inside the container stays 9999; only the host-facing port changes.

## 0.6 localhost, LAN addresses, and Docker networking

- localhost always means "this machine" from the point of view of the program using the URL.
- If Jellyfin is installed directly on the same host as M3U Web Picker, localhost:9999 is usually fine.
- If Jellyfin is on another computer, use the LAN IP or hostname of the computer running M3U Web Picker, for example `http://192.168.1.50:9999`.
- If Jellyfin itself runs inside another Docker container, localhost refers to the Jellyfin container, not the M3U Web Picker container. The safest beginner option is usually the Docker host LAN IP. A shared Docker network/service-name setup is an advanced alternative.
- If another device cannot reach port 9999, check the host firewall before changing the application.

### **Do not port-forward this to the public internet by default**

The app manages provider credentials, stream routes, guide configuration, and local automation state. Treat it as a LAN / trusted-network application unless you intentionally add authentication and a secure reverse proxy.

## 0.7 Persistent state and backups

The normal Compose stack stores live application state in the Docker named volume `m3u-picker-data`. Rebuilding the image or force-recreating the container does not normally erase that volume.

Nightly SQLite backups are written to the host backup folder mapped to `./backups` by default. The Compose file currently schedules backups around 3:15 AM America/New\_York and retains 30 days. You can point `M3U_BACKUP_DIR` at another host folder in `.env`.

```
# .env example
M3U_BACKUP_DIR=./backups
M3U_HOST_PORT=9999
```

### **Do not confuse container deletion with volume deletion**

`docker compose down` leaves the named volume in place. Commands that explicitly delete volumes (for example `docker compose down -v`) remove persistent application state. Do not use `-v` unless you intentionally want a factory reset.

# 1. Top Bar - Playlist Output and Master Update



*The top of the app contains the playlist URL, Copy / Manage Order controls, and the application-wide Master Update schedule.*

The top bar is the destination of the whole application: a media client ultimately needs the channel playlist and the EPG. RC2 currently shows the playlist URL here; the planned next UI cleanup also moves the single EPG URL here.

## 1.1 Playlist URL

RC2 currently serves the normal playlist at:

```
/playlist/custom.m3u
```

Planned user-facing name:

```
/playlist/channels.m3u
```

### Localhost and the Copy button

Same machine: `http://localhost:9999/playlist/custom.m3u`. 'localhost' always means the machine making the request. If Jellyfin/Plex runs on another computer, use the M3U Web Picker computer's IP or hostname instead, for example `http://192.168.1.50:9999/playlist/custom.m3u`. The Copy button uses the host and port you opened.

## 1.2 Master Update

Master Update owns the complete refresh cycle. There is intentionally no separate hourly sports schedule or independent guide scheduler in the normal workflow.

- Automatic update is enabled by default.
- The default time is 3:00 AM in the application / selected local context.
- Changing the time changes the next scheduled run.
- Update Now runs the same pipeline immediately.
- A manual Update Now does not move the next normal daily run.
- While running, the UI shows a spinner and an elapsed-time counter.
- The last completed cycle records how long it took.



*A manual update shows a live elapsed timer. Long provider/EPG scans can legitimately take several minutes.*

### First runs can be slow

Provider parsing and public EPG filtering dominate run time. A multi-minute update is not automatically a failure. Watch the running status instead of repeatedly clicking Update Now.

## 2. Provider Sources

The primary provider is the only catalog shown in Channel Manager. Ordered fallback providers are used only by Sports Automation when a usable primary feed is missing.

| Primary name | Provider or M3U URL          | Xstream username | Xstream password |
|--------------|------------------------------|------------------|------------------|
| Primary      | https://provider.example.com | Optional         | Optional         |

For Xstream, enter only the server/base URL, and use the separate username/password fields. The app probes `player_api.php`, then builds the playlist and XMLTV endpoints internally.

**M3U File**

Choose File No file chosen

*Primary provider entry supports either a provider/server URL plus optional Xstream credentials, or an uploaded M3U file.*

Provider Sources is where playable channel streams come from. The app supports one Primary source and optional ordered fallback sources for Sports Automation.

### 2.1 Primary provider - the normal channel catalog

The Primary provider is the only provider catalog shown in Channel Manager. This prevents fallback services from flooding the manual channel list with duplicates.

### 2.2 Common input formats

| What your provider gave you           | What to enter                                                                                    |
|---------------------------------------|--------------------------------------------------------------------------------------------------|
| Server/base URL + username + password | Enter the base URL in Provider or M3U URL and use the separate Xstream username/password fields. |
| One long M3U URL                      | Paste the full M3U URL into Provider or M3U URL. Embedded credentials remain supported.          |
| A local .m3u/.m3u8 file               | Use M3U File and Use File as Primary. This is a static snapshot until you upload a newer file.   |

#### Xstream users

For Xstream, prefer the base/server URL plus the separate username and password boxes. The app probes `player_api.php`, imports live streams/categories only, and derives the provider guide endpoint internally.

### 2.3 Loading and replacing a Primary source

- Only one Primary source can exist at a time.
- After a Primary provider is loaded, its credential/input controls are intentionally disabled to prevent accidental mutation.
- To change providers or credentials, remove the current Primary first, then add the replacement.
- Manual saved-channel selections are associated with the Primary catalog. Replacing the provider can change what those selections resolve to; test carefully.

## 2.4 Very large providers

RC2 includes import safeguards because some providers expose enormous live/VOD/series catalogs. The normal Compose defaults include:

| Safeguard                      | Default                      |
|--------------------------------|------------------------------|
| Maximum provider live channels | 50,000                       |
| Provider warning threshold     | 20,000 channels              |
| Maximum provider playlist size | 100,663,296 bytes (~96 MiB)  |
| Maximum provider JSON size     | 100,663,296 bytes (~96 MiB)  |
| Maximum compressed public EPG  | 268,435,456 bytes (~256 MiB) |

The first four can be overridden in .env using the M3U\_MAX\_PROVIDER\_\* variables included in .env.example. Raising limits also raises memory/CPU risk; do not increase them just to silence a bad provider export.

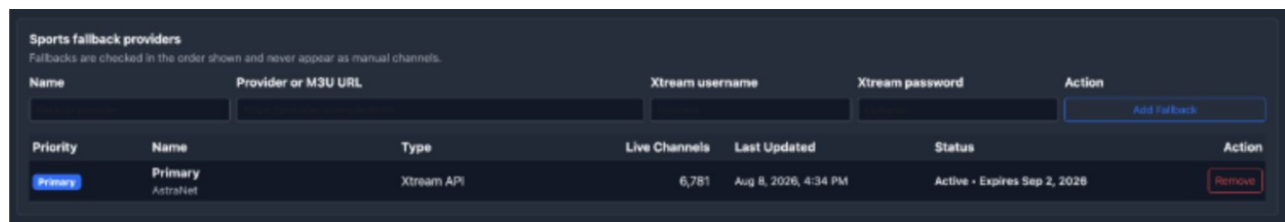
### Prefer live-only Xstream imports

The app deliberately requests live streams/categories instead of blindly downloading VOD and series catalogs. This is important for providers with hundreds of thousands of non-live entries.

## 2.5 Credentials and privacy

- Credentials are not shown back in the normal channel table or provider rows.
- Separate credential fields reduce accidental browser/API exposure, but they are not encryption.
- Provider credentials and API secrets live in local application configuration/state. Protect the Docker data volume and backup folder.
- Exported playlists and provider stream URLs may themselves contain sensitive tokens or credentials. Treat M3U files as secrets if your provider encodes credentials in stream URLs.

## 3. Sports Fallback Providers



*Fallback providers are ordered and used by Sports Automation only when a usable higher-priority feed is missing.*

### Most first-time users should skip this

If you have one IPTV subscription, leave the fallback section empty.

A fallback is not a second manual-channel catalog. It is a sports rescue source. During a sports scan, the app checks providers in priority order and uses a fallback only when no usable higher-priority candidate can satisfy the event.

- Fallback channels never appear in the normal Channel Manager catalog.
- A fallback is registered with lightweight validation; full scanning happens during the master sports cycle.

- Use the Use backup feeds option in Sports Automation to allow or disallow fallback feeds.
- A lower-priority provider should not create extra duplicate sports feeds when the Primary already provides a usable candidate.
- If a huge fallback provider causes resource problems, remove it or reduce the provider rather than assuming every provider must be loaded.

## 4. EPG Manager (RC2) - and the guide-output transition



**EPG Manager**  
Built-in and external XMLTV guides served by this application. Stored provider URLs and credentials are hidden after saving.

| NAME                         | TYPE     | SERVED URL                              | STATUS                                      | ACTION  |
|------------------------------|----------|-----------------------------------------|---------------------------------------------|---------|
| <input type="text"/>         | External | <input type="text"/>                    |                                             | Add     |
| Combined                     | Built-in | http://localhost:10000/epg/combined.xml | Copy Updated Aug 8, 2026, 4:41 PM<br>222 KB | Managed |
| Sports                       | Built-in | http://localhost:10000/epg/sports.xml   | Copy Updated Aug 8, 2026, 4:41 PM<br>64 KB  | Managed |
| No additional XMLTV sources. |          |                                         |                                             |         |

*RC2 EPG Manager. The product direction is to simplify this so the normal client-facing guide is simply called EPG.*

EPG means Electronic Program Guide. In this app, EPG sources are inputs; the app filters/reconciles them and publishes one final guide for the client.

### 4.1 RC2 terminology

Current RC2 final guide: /epg/combined.xml  
Planned public name: /epg/epg.xml

"Combined" is legacy terminology. The intended product concept is simply EPG: provider guide data, configured guide data, public-country fallback data, and generated sports guide data all end up in one client-facing guide.

#### Planned UI simplification

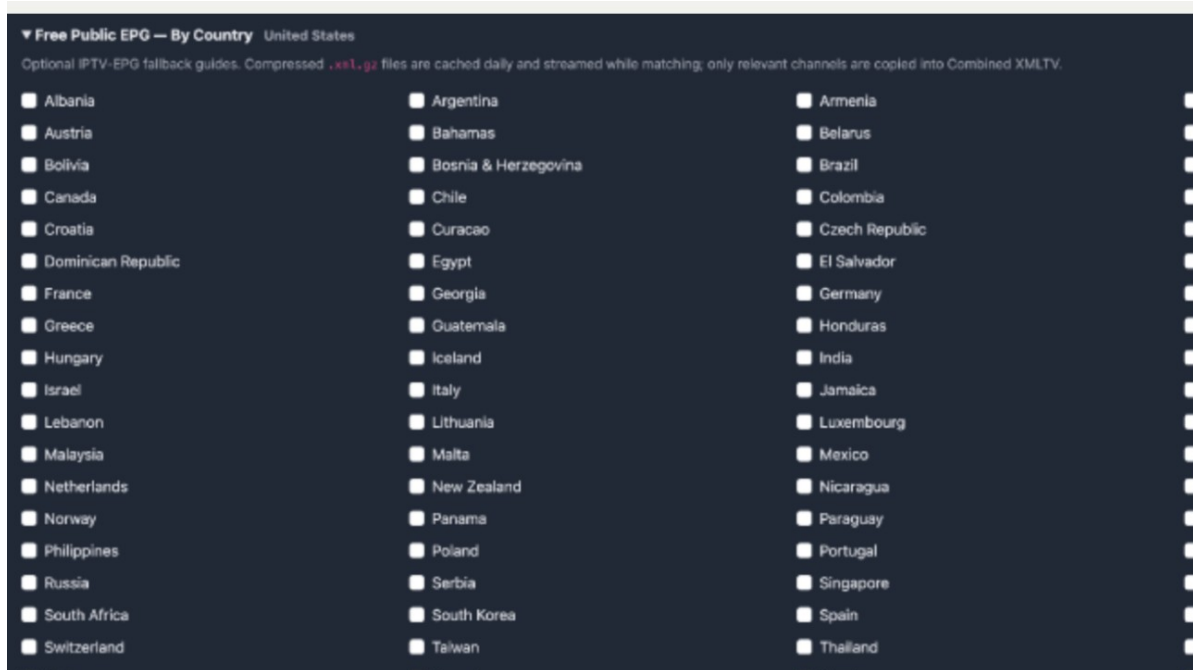
The current EPG Manager is expected to shrink. The EPG link will move to the top beside the playlist URL, and the normal EPG section will mainly keep the worldwide country selector. This guide explains RC2 as shipped and the intended direction so testers are not surprised by the next cleanup.

### 4.2 External XMLTV sources

Advanced users can add external XMLTV sources. For ordinary setups, provider guide discovery plus the public country guide is usually enough. If you add an external guide, use a meaningful name and avoid adding the exact same guide again directly in Jellyfin; let M3U Web Picker own reconciliation.



## 5. Free Public EPG - By Country



*The public EPG country selector is a general guide fallback, not a sports-only feature.*

This section downloads free compressed country guides and uses them to fill missing guide windows for any selected channel - local stations, cable networks, sports networks, and generated sports matching candidates.

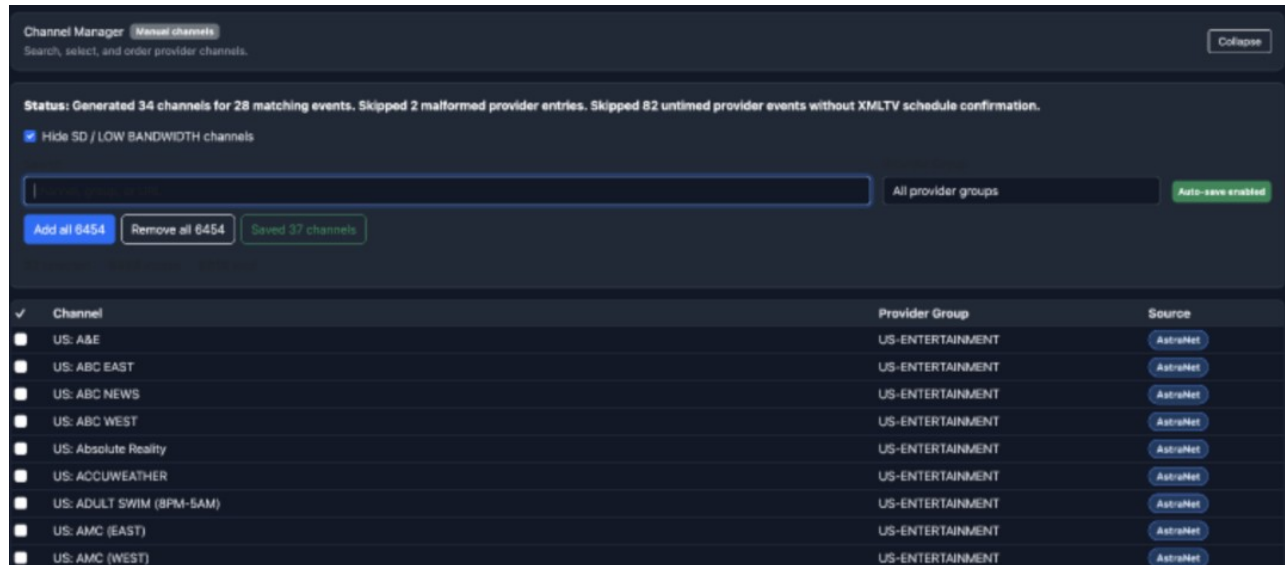
- United States is enabled by default on a fresh setup.
- Enable only countries you actually need. Unchecked countries are not downloaded.
- The app downloads .xml.gz files and keeps them compressed on disk.
- Large guides are stream-decompressed and filtered; the app does not intentionally expand a giant worldwide XML file into permanent storage.
- Provider/configured guide data has higher priority. Public guide data fills uncovered time windows rather than blindly replacing good provider data.
- Public guide data can also corroborate sports matching, but it does not become authoritative over a strong schedule API + provider match.

Public guide source: <https://iptv-epg.org/guides>

### Why not add the giant public guide directly to Jellyfin?

Because M3U Web Picker can filter it down to only the channels/programs you actually use. Jellyfin then receives one much smaller, reconciled guide instead of several overlapping guide sources.

## 6. Channel Manager



*Channel Manager contains manually selected Primary-provider channels plus read-only generated sports channels.*

Channel Manager is where you build the stable part of your lineup.

### 6.1 Manual channels vs generated sports channels

| Type                     | Meaning                                                     | Lifecycle                                                      |
|--------------------------|-------------------------------------------------------------|----------------------------------------------------------------|
| Manual / saved channel   | You explicitly selected it from the Primary catalog.        | Stays until you remove it.                                     |
| Generated sports channel | Sports Automation created it for a scheduled/matched event. | Appears/disappears with the event window and automation state. |

#### Hard rule

Sports Automation must never delete, replace, suppress, or hijack a manually saved channel - even if both ultimately point at the same provider stream.

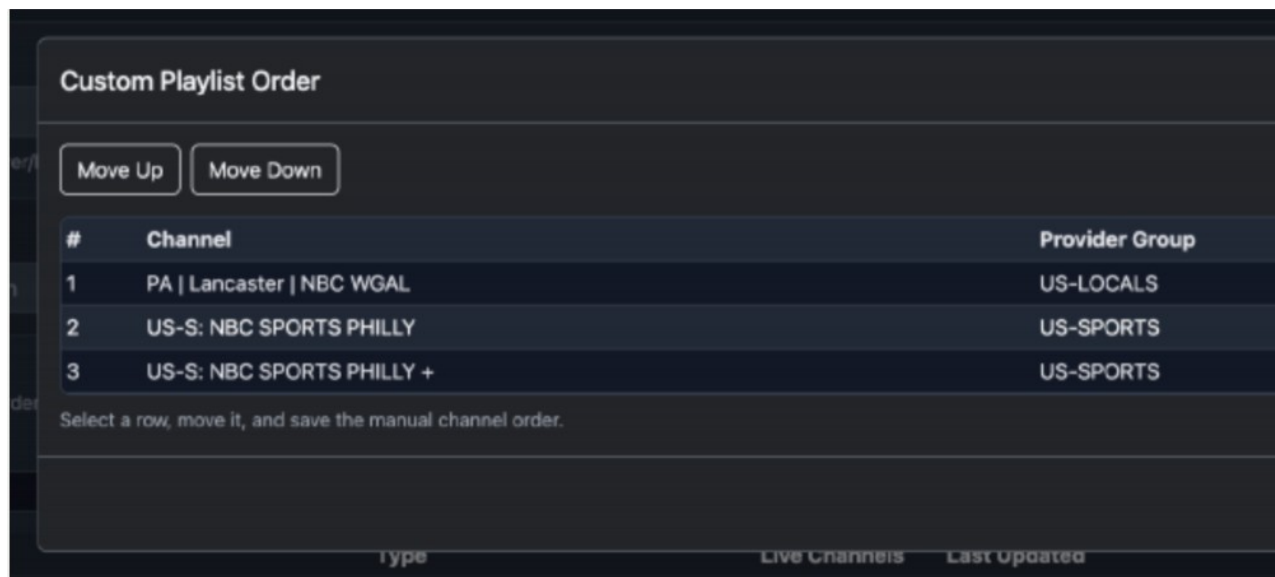
### 6.2 Search and filtering

- Use search to narrow a large provider list.
- Provider Group filters help when the provider has useful categories.
- Hide SD / low-bandwidth can reduce clutter and also affects sports feed eligibility.
- Manual selections auto-save; there is no separate "Save channels" button in the normal flow.
- Generated sports rows are read-only. Change Sports Automation rules rather than trying to edit those rows manually.

### 6.3 Provider catalog changes

If the provider renames/removes/reorders channels, saved selections are reapplied against the refreshed Primary catalog. If a formerly selected provider channel disappears entirely, the app cannot tune it until the provider offers it again or you select a replacement.

## 7. Manage Order - Manual Channels Only



*Manage Order changes only the saved/manual channel order.*

Use Manage Order if you care about the order/numbering of stable manual channels. Sports channels use automation numbering blocks and are not dragged around manually.

### **Do not try to make sports numbering behave like manual numbering**

Sports channels are transient and must remain stable inside their competition blocks for a given event. Manual ordering and sports allocation are intentionally separate systems.

## 8. Sports Automation - Enable / Disable

Sports Automation is optional. If you only want a curated IPTV playlist and a merged guide, leave Sports Automation off; Sections 0-7 still work.

- Enable sports: generated sports channels participate in the final playlist/guide.
- Disable sports: generated rows are immediately hidden from served outputs.
- RC2 keeps disabled generated data privately cached for up to 24 hours so an accidental toggle can be reversed quickly.
- Your saved sport/league/team rules are preserved when sports is disabled.

## 9. Sports Channel Block

This subsection controls how generated channels are numbered and which schedule window is considered.

| Control            | What it means                                   | Beginner recommendation                                         |
|--------------------|-------------------------------------------------|-----------------------------------------------------------------|
| First league block | Base number used for the competition block map. | Leave 1000 unless it conflicts with your manual numbering plan. |
| Channels per event | Number of numbered slots reserved per event.    | Leave the default unless you know you need more feed            |

| Control      | What it means                                          | Beginner recommendation                                                       |
|--------------|--------------------------------------------------------|-------------------------------------------------------------------------------|
|              |                                                        | variants.                                                                     |
| Group title  | Playlist group/category for generated sports channels. | Use a name that is obvious in your client.                                    |
| Timezone     | Timezone used to interpret sports dates/times.         | Set this to where you actually watch / schedule games.                        |
| Event window | Today, today+tomorrow, or next 24 hours.               | Today is simplest; use next 24 hours if overnight/timezone boundaries matter. |

## 9.1 Numbering collisions

Sports competitions receive separate 1,000-channel blocks. If manual channel numbering overlaps the configured sports start, the application should preserve manual channels and move sports allocation rather than overwriting them.

## 9.2 Midnight and long-game edge cases

- Games crossing midnight should remain the same logical event; midnight is not a reason to create a second game.
- Generated channels receive a 90-minute postgame grace window so long games and overtime are not removed too early.
- For college football / other long events, the grace period is intentionally conservative.
- A later rebroadcast is conceptually a replay of the original event, not a new event. RC2 still has a known replay-suppression defect in this area.

## 10. Schedule API

The screenshot shows the 'Sports Automation' interface with a dark theme. At the top, it says 'Sports Automation Enabled' and '2 selections • 34 generated channels'. Below this is a 'Sports channel block' section with a description: 'Generated channels stay separate from normal channel numbering.' It contains three input fields: 'First league block' with the value '1000', 'Channels per event' with the value '10', and 'Group title' with the value 'Sports Today'. Below these are 'Timezone' set to 'America/New\_York' and 'Event window' set to 'Current sports day'. A note states: 'Each league/series gets 1,000 channels; 100 event slots at 10 channels per event. Overflow uses a separate continuation block.' There is a button 'View league / series channel blocks'. The 'Schedule API' section follows, with a description: 'Optional canonical schedule data. Leave this off to use the existing provider-derived matching.' It has an 'API URL' field (containing a placeholder) and an 'API key' field (containing a placeholder). A message says: 'Schedule API enabled but not fully configured - enter the API URL and key, then save.' Below this is a table titled 'Loaded schedule APIs'.

| API          | Scope | URL | Status     | Last Update |
|--------------|-------|-----|------------|-------------|
| API-BASEBALL | MLB   | —   | Incomplete | —           |

At the bottom, a note reads: 'The master application update runs in dependency order: schedule cache → provider/channel refresh → guide refresh → sports match/build → M3U/XMLTV publish.'

*Schedule API is optional. When disabled, the API fields are disabled and provider-derived matching remains active.*

The schedule API gives Sports Automation an authoritative list of games and stable event IDs before it looks through provider channel names and guide data.

### 10.1 Why it matters

Without a schedule API, the app must infer that text such as "Phillies vs Blue Jays" refers to the correct live game. With the API, it first knows the scheduled matchup/time and can score provider/guide candidates against that game.

#### Think of it as the game calendar, not the video source

The API does not stream the game. It tells the app which event exists; the IPTV provider still supplies the playable feed.

### 10.2 RC2 implementation status

- The schedule API is optional.
- RC2 has a tested API-SPORTS Baseball/MLB path. The UI/model is being generalized for additional sport-specific API products later.
- API-SPORTS uses sport-specific products/endpoints; there is not one universal base URL for every sport.
- The Baseball endpoint tested during RC2 is <https://v1.baseball.api-sports.io>.

- The app fetches the date + timezone response and filters MLB locally.
- API secrets are stored server-side and are not returned to the browser after saving.

API-SPORTS dashboard: <https://dashboard.api-football.com/>

### 10.3 API off / blank / failure behavior

- If the switch is off, the API configuration is not effective and canonical API cache data is ignored for matching.
- If URL/key configuration is incomplete, the app falls back to provider-derived matching.
- If a configured API fetch fails or is over quota, a valid cached schedule may still be used when available.
- If no usable canonical schedule is available, the scan should continue using the legacy provider/EPG matcher instead of taking the lineup down.
- Do not repeatedly force API refreshes just because a provider channel is missing; quota and provider availability are separate problems.

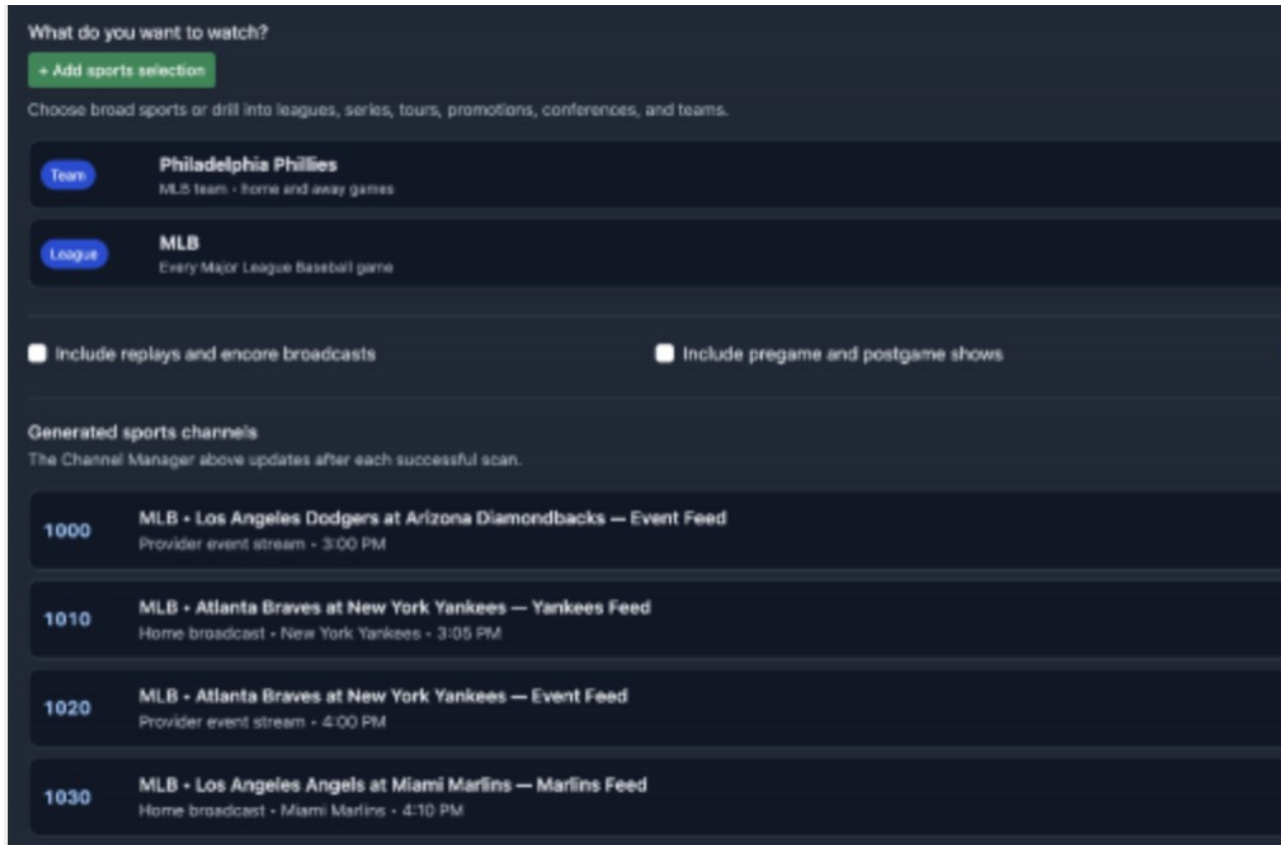
### 10.4 Expected canonical behavior

```
API game: Blue Jays @ Phillies, 6:05 PM
  6:00 PM betting / gameday show -> supporting content, not the game
  6:05 PM live full game         -> canonical live airing
  11:00 PM rebroadcast           -> replay of the same API event
```

#### Current RC2 caveat

The intended behavior above is not fully enforced yet in the served result: late Phillies rebroadcasts can still leak through. Use this example as a tester expectation and bug-report reference.

## 11. What Do You Want to Watch?



*Sports selections can be broad (sport/league) or specific (team/competitor).*

This is where you define sports rules. The picker supports Sport -> League/series/tour/promotion/division -> Conference -> Team/competitor.

### 11.1 Broad league rule

Baseball -> MLB -> All teams

Expected result: one best feed per MLB game unless another explicit rule says you care more about a particular team/event.

### 11.2 Explicit team rule

Baseball -> MLB -> Philadelphia Phillies

Expected result: Phillies games may expand into a fuller feed set (for example home/away/national/event) while other league games remain one-best-feed.

### 11.3 League + team together

|                              |                                   |
|------------------------------|-----------------------------------|
| 15 MLB games today           |                                   |
| MLB selected broadly         | -> 1 best feed for ordinary games |
| Phillies selected explicitly | -> expanded Phillies feed set     |
| Same Phillies game           | -> one logical event, not two     |

### Selections overlap on purpose

Selecting MLB plus Phillies is not redundant. The broad MLB rule gives coverage; the explicit Phillies rule expresses feed preference for Phillies games.

## 11.4 Everything Mode

Everything Mode scans every detected event while preserving your curated selections. It is useful for stress testing and discovery, but it can make scans much larger and noisier.

### Beginner recommendation

Leave Everything Mode off. Start with one league and one team, verify the result, then expand.

## 11.5 Cross-sport and naming ambiguity

Provider text can contain generic mascots such as Eagles, Tigers, or Giants across unrelated sports. Matching is designed to be sport/league-aware so a baseball or foreign-team "Eagles" should not satisfy an NFL Philadelphia Eagles rule. If you see a cross-sport false match, capture the generated channel name plus provider/guide text and [report it](#).

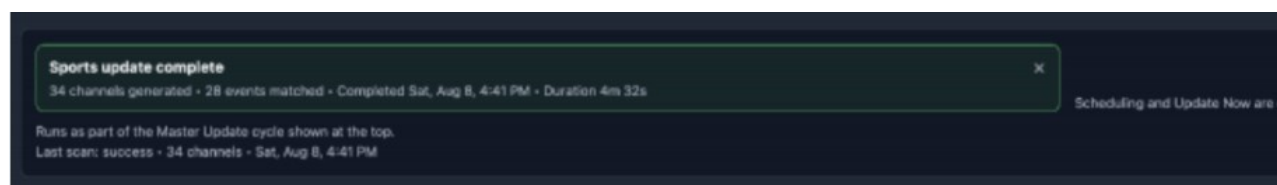
## 12. Sports Options - Replays, Pregame/Postgame, Backups

| Option                                          | When OFF                                                            | When ON                                                                                                   |
|-------------------------------------------------|---------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| Include replays and encore broadcasts           | Only the primary live airing should be represented.                 | Later same-event airings may be represented as replay windows on the same logical event/channel identity. |
| Include pregame and postgame shows              | Studio/pregame/postgame rows should not become separate game feeds. | Supporting shows may be included where matching logic supports them.                                      |
| Use backup feeds when primary feeds are missing | Only Primary-provider sports candidates are eligible.               | Ordered fallback providers can satisfy missing feeds.                                                     |

### Replay bug in RC2

If replays are OFF and a late-night copy of the same game appears as another generated channel/feed, [report it](#). That is not intended behavior.

## 13. Generated Sports Channels and Status



*Sports status is currently shown at the bottom; the longer-term UI direction is to rely more heavily on the top Master Update status.*



After a successful scan, generated channels appear in the Sports Automation preview and in Channel Manager. Each generated channel receives a credential-free synthetic ID tied to its assigned channel slot, preventing Jellyfin from collapsing generated feeds that happen to share an underlying provider channel.

```
Example generated tvg-id:  
m3u-picker-sports-1070
```

- The same synthetic ID is used in the M3U channel and XMLTV channel/program records.
- Generated channels are temporary; stale events are removed on successful updates after their event/grace window.
- A failed scan preserves the last-known-good generated lineup rather than publishing half-built output.
- A successful scan with zero matches clears stale sports output.
- The status area reports success/failure, generated counts, timing, and current update state.

## 14. What Update Now Actually Does

The app runs one dependency-ordered pipeline. This is the internal sequence behind the simple Update Now button:

| Order | Stage                                    | Purpose                                                                     |
|-------|------------------------------------------|-----------------------------------------------------------------------------|
| 1     | Schedule API (optional)                  | Refresh/corroborate canonical events if enabled and due.                    |
| 2     | Primary/fallback provider refresh        | Refresh channel catalogs that are eligible for the scan.                    |
| 3     | Provider/configured/public guide refresh | Update guide inputs; public-country sources refresh from compressed caches. |
| 4     | Sports event matching                    | Map provider/guide candidates to canonical or provider-derived events.      |
| 5     | Feed selection and channel build         | Apply league/team rules, provider priority, options, and numbering.         |
| 6     | EPG reconciliation                       | Build the final filtered guide plus generated sports programs.              |
| 7     | Playlist build                           | Build the final selected/manual + generated channel playlist.               |
| 8     | Validation and atomic publish            | Validate expected outputs and replace published files as a unit.            |

### Why this order matters

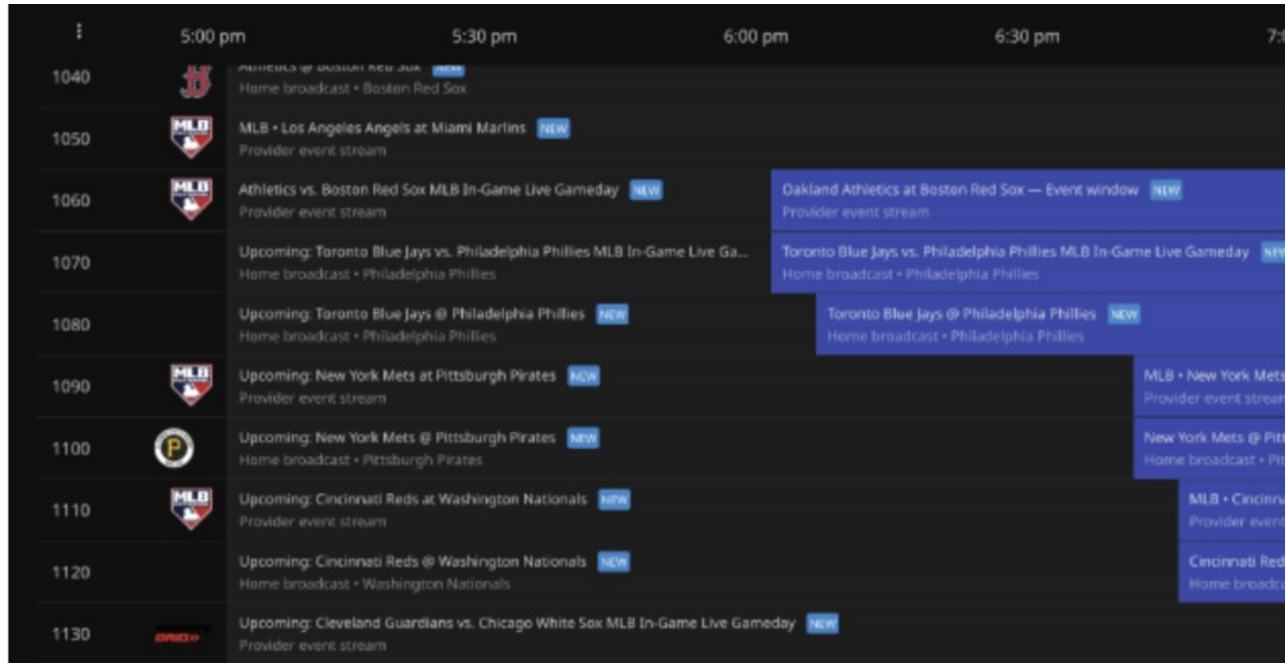
Sports matching depends on schedules/providers/guides being current. Publishing comes last so Jellyfin does not see a playlist from one scan paired with a half-built guide from another.

### 14.1 Failure behavior

- A failed provider refresh or sports scan should preserve previously published working outputs.
- Public EPG failure should not automatically destroy provider guide data.

- Schedule API failure should degrade to cache/legacy matching instead of taking television down.
- If the application process is restarted during a scan, stale running-state handling should report an interrupted failure rather than remaining permanently "running".

## 15. Connect Jellyfin



*Example generated sports channels and guide windows in Jellyfin.*

Jellyfin should ultimately need exactly two application outputs: the playlist and the EPG.

### 15.1 RC2 URLs

Tuner / M3U:  
<http://YOUR-SERVER-IP:9999/playlist/custom.m3u>

TV Guide / XMLTV:  
<http://YOUR-SERVER-IP:9999/epg/combined.xml>

### 15.2 Planned names after the cleanup

Tuner / M3U:  
<http://YOUR-SERVER-IP:9999/playlist/channels.m3u>

TV Guide / EPG:  
<http://YOUR-SERVER-IP:9999/epg/epg.xml>

#### Only use two M3U Web Picker outputs in Jellyfin

Do not also add sports.xml and the raw public USA guide to Jellyfin. The app is supposed to own reconciliation and hand Jellyfin one final guide.

### 15.3 Jellyfin step-by-step

1. Open Jellyfin Dashboard -> Live TV.

2. Add an M3U tuner and paste the playlist URL from M3U Web Picker.
3. Add an XMLTV / TV Guide data provider and paste the M3U Web Picker EPG URL.
4. Save the tuner and guide source.
5. Run Refresh Guide Data.
6. Verify one manual channel first, then verify generated sports channels.

## 15.4 Jellyfin caching problems

### Jellyfin caches aggressively

During RC testing, stale tuner/guide mappings can survive URL/content changes. If the app output looks correct but Jellyfin still shows impossible old channels or guide rows: remove and re-add the affected tuner/guide source, restart Jellyfin if needed, then refresh guide data.

- Do not assume every stale channel is an M3U Web Picker bug until you compare the served M3U/XMLTV directly.
- If multiple XMLTV sources describe the same channel, Jellyfin may choose unexpectedly. This is another reason to keep only the app EPG configured.
- If Jellyfin is in Docker, remember the localhost rule from Section 0.6.

## 16. Other Clients / Plex Notes

Any client that directly accepts M3U plus XMLTV can use the same two conceptual outputs. Plex is more particular about tuner discovery and may require an intermediary/tuner bridge depending on your setup; this RC2 guide does not claim direct Plex M3U support. Validate your Plex path separately rather than assuming Jellyfin instructions map one-for-one.

### VLC sanity check

If you only need to prove that a playlist stream works, opening the playlist or an individual stream in VLC is a useful way to separate "provider stream is dead" from "media-server guide/tuner configuration is wrong."

## 17. Updating, Restarting, and Backups

### 17.1 Restart the existing container

```
docker compose up -d
```

### 17.2 Stop the stack

```
docker compose down
```

### 17.3 Pull new code and rebuild

```
git status
git pull
docker compose up -d --build
```

### If git status is not clean

Do not blindly git pull over local modifications. Either commit/stash intentional changes or make a clean clone. Release/testing work is much easier when the repo is a clean source of truth and runtime data lives only in Docker volumes/backups.

## 17.4 Manual database backup

```
docker compose exec -T m3u-picker python /app/backup_db.py
```

## 17.5 Factory reset warning

```
# DANGEROUS: removes the named data volume
docker compose down -v
```

Only use that when you intentionally want to erase saved configuration/database state. A normal down/up or rebuild does not require -v.

# 18. Troubleshooting by Symptom

| Symptom                                        | What to check first                                                                                                                                  |
|------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| Web page will not open                         | Confirm container is running: <code>docker compose ps</code> . Check docker compose logs. Verify host port and firewall.                             |
| Port 9999 already allocated                    | Set <code>M3U_HOST_PORT</code> in <code>.env</code> , then recreate the stack. Use the new host port in Jellyfin.                                    |
| Jellyfin cannot reach app                      | Do not use localhost from another machine/container. Use the Docker host LAN IP/hostname and check firewall.                                         |
| Primary provider will not load                 | Recheck URL/credentials. For Xstream use base URL + separate username/password. Very large catalogs may hit import safeguards.                       |
| Provider inputs are disabled                   | This is expected after a Primary is loaded. Remove the Primary before changing credentials/source.                                                   |
| Uploaded M3U never refreshes                   | A file Primary is a static snapshot. Upload a newer file; only URL/Xstream sources can be refreshed from the provider.                               |
| Channel missing from Channel Manager           | Only the Primary catalog appears there. Fallback providers never populate manual channels. Check provider group/search/SD filters.                   |
| Manual channel disappeared after sports update | That is a bug. Manual/static selections must survive sports dedupe and may coexist with sports channels using the same underlying stream.            |
| Sports game missing                            | Verify the game is inside the configured event window/timezone; check selection rule, provider candidates, SD filter, backup setting, API status and |

| Symptom                             | What to check first                                                                                                                                                                                                  |
|-------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                     | guide data.                                                                                                                                                                                                          |
| Too many sports channels            | Look for overlapping league/team selections, replays, pre/postgame, Everything Mode, or fuzzy provider events. Team selection should expand only the selected team game; broad league games should be one best feed. |
| Midnight replay shows as new game   | Known RC2 replay-suppression defect. Replays OFF should prevent this; capture channel IDs/times and report it.                                                                                                       |
| Schedule API says disabled          | Turn Use schedule API on and save a valid URL/key. Fields are disabled while the switch is off.                                                                                                                      |
| Schedule API fetch fails            | The app should use valid cache when available or fall back to provider-derived matching. Check API dashboard/quota separately.                                                                                       |
| Guide missing for normal channel    | Provider guide wins; public-country EPG may fill gaps. Make sure the relevant country is enabled and the channel ID/name can be matched.                                                                             |
| Guide is stale in Jellyfin          | Refresh Guide Data. If still stale, remove/re-add tuner or XMLTV source, restart Jellyfin, refresh again.                                                                                                            |
| Update takes several minutes        | Normal for provider/XMLTV parsing. Watch spinner/stage/elapsed time. Avoid repeatedly launching updates.                                                                                                             |
| Update fails but old lineup remains | That is expected last-known-good behavior. Inspect logs/status to fix the failed input, then run Update Now again.                                                                                                   |
| Docker rebuild lost settings        | A normal rebuild should not remove the named data volume. Check whether someone ran down -v or changed Compose project/volume names.                                                                                 |
| Backup folder is empty              | Check bind path M3U_BACKUP_DIR, filesystem permissions, host timezone assumptions, and that the container has been running at backup time.                                                                           |
| Memory use explodes                 | Huge provider/public guides are the usual cause. Keep live-only provider imports, avoid giant fallback catalogs, and do not raise safety limits casually.                                                            |

## 18.1 Useful diagnostic commands

```
docker compose ps
docker compose logs --tail=200 m3u-picker
docker stats --no-stream m3u-picker
```

To inspect the two client-facing outputs directly in RC2:

```
curl -sS http://localhost:9999/playlist/custom.m3u -o ~/Desktop/channels.m3u
curl -sS http://localhost:9999/epg/combined.xml -o ~/Desktop/epg.xml
```

### Credential warning before sharing M3U files

A provider M3U can contain credentials/tokens in stream URLs. Sanitize it before posting publicly or sending it to someone you do not trust.

## 19. Known RC2 Limitations / Tester Expectations

| Area                   | Intended behavior                                                      | RC2 status / tester note                                                                                   |
|------------------------|------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| Replay suppression     | Later same-game rebroadcasts should be ignored when replays are OFF.   | Still imperfect. Midnight/late Phillies rebroadcasts have been observed.                                   |
| Public output names    | Simple /playlist/channels.m3u and /epg/epg.xml.                        | Not yet renamed in RC2; current routes are custom.m3u and combined.xml.                                    |
| EPG UI                 | One top EPG link plus worldwide country selection for normal users.    | RC2 still shows EPG Manager. Simplification is planned.                                                    |
| Sticky top controls    | Playlist/EPG/master-update controls remain accessible while scrolling. | Planned, not yet built.                                                                                    |
| Schedule APIs by sport | Adapter-based sport-specific schedule APIs.                            | Baseball/MLB is the tested pilot; additional sports need adapter/testing work.                             |
| Jellyfin cache         | Client refresh reflects served files.                                  | Jellyfin can retain stale tuner/guide mappings during rapid RC changes; manual source reset may be needed. |

### How to report a useful sports bug

Include the selected rules, timezone/event window, generated channel name/number, expected game time, whether replays/pregame/backups were enabled, and (if safe) sanitized provider/EPG snippets. Stable API event IDs are especially useful when available.

## 20. Security and Privacy

- Treat provider credentials, API keys, the Docker data volume, backup databases, and raw provider M3U files as secrets.
- The UI intentionally hides stored provider credentials after save; that does not make the underlying local configuration encrypted.
- Do not expose port 9999 directly to the public internet unless you add an appropriate security layer.
- Generated sports IDs are credential-free, but underlying playback still ultimately reaches provider streams.
- When asking for help, prefer screenshots of UI/status plus sanitized M3U/EPG excerpts rather than raw credential-bearing files.

- If you publish logs, scan them first. Tip: search for http, user, pass, token, key, and x-apisports-key; replace secret values or credential-bearing URLs with [REDACTED] before sharing.

## 21. Glossary - enough IPTV vocabulary to use the app

| Term                     | Plain-English meaning                                                                                             |
|--------------------------|-------------------------------------------------------------------------------------------------------------------|
| M3U                      | A text playlist describing channels and the URLs used to play them.                                               |
| EPG                      | Electronic Program Guide - the schedule/listings shown in a TV client.                                            |
| XMLTV                    | A common XML file format used to deliver EPG data.                                                                |
| Xtream                   | A provider login/API style that commonly uses a server URL plus username/password.                                |
| Primary provider         | The IPTV source that supplies the normal Channel Manager catalog and first-choice sports streams.                 |
| Fallback provider        | Optional sports-only backup source used when a higher-priority provider cannot satisfy an event.                  |
| Manual channel           | A normal provider channel you explicitly saved.                                                                   |
| Generated sports channel | A temporary channel created by Sports Automation for an event/feed.                                               |
| tvb-id                   | The identifier used to connect an M3U channel to its EPG channel record.                                          |
| Canonical event          | The app's single logical identity for one actual game, ideally anchored by a stable schedule API event ID.        |
| Replay / encore          | A later airing of an already-known event, not a new game.                                                         |
| Public EPG               | Free country guide data used as a low-priority fallback/enrichment source.                                        |
| Master Update            | The single daily/manual pipeline that refreshes schedules, providers, guides, sports matching, playlist, and EPG. |
| Last-known-good          | Previously published output kept in place when a new update fails before safe publication.                        |

## 22. Final Setup Checklist

- Docker is running and the app opens on the expected host/port.
- Exactly one Primary provider is loaded.
- The normal channels you want are saved in Channel Manager.
- Relevant public EPG countries are enabled; United States is the default.
- Sports Automation is either intentionally OFF or has a small, understandable set of rules.
- Schedule API is either intentionally OFF or shows a valid configured/loaded status.
- Timezone and event window match how you want sports days interpreted.
- Replay/pregame/backup options are set deliberately.

- Update Now completes successfully and the elapsed/status result looks reasonable.
- Jellyfin has only the M3U Web Picker playlist plus the M3U Web Picker EPG for this tuner flow.
- A manual channel tunes and has guide data.
- A generated sports channel (if enabled) tunes and has the expected guide event.
- You know where backups are stored and have not exposed the management port publicly.

#### Done means done

If your selected channels and generated sports feeds appear in Jellyfin with sane guide data, stop configuring things. The advanced knobs exist for edge cases; they are not a checklist of features every user needs to turn on.

## Useful Links

**Project sports branch:** <https://github.com/zschmook/m3u-web-picker/tree/sports>

**Docker Desktop:** <https://www.docker.com/products/docker-desktop/>

**Git downloads:** <https://git-scm.com/downloads>

**API-SPORTS dashboard:** <https://dashboard.api-football.com/>

**Public EPG guide list:** <https://iptv-epg.org/guides>